

JET FIRE RATED PROTECTION

CHARCOAT JF

INTUMESCENT JET FIRE RATED COATING
FOR THE FIRE PROTECTION OF VARIOUS
SUBSTRATES



CHARCOAT

GENERAL INFORMATION

CharCoat JF is our revolutionary new Hydrocarbon Jet Fire Rated Coating which is a 2 component 100% Epoxy-based Intumescent, 100% Solvents-free.

CharCoat JF is suitable for the protection of steel, aluminium or composites from Cellulosic to Hydrocarbon Pool and Jet Fires.

CharCoat JF is intended to protect Structural Steel, vessels, piping, actuators, valves, piping and cable trays from hydrocarbon pool and/or jet fire to extend the functional integrity for an extended period of time.

As tested by Intertek, CharCoat JF can withstand 1400°C constant temperature under ISO 22899-1 Jet Fire Standard, far and above the standard hydrocarbon pool fire (1100°C). To typically describe a Jet Fire; a jet or spray fire is a turbulent diffusion flame resulting from the combustion of a fuel continuously released with some significant momentum in a particular direction or directions.”

APPLICATION

CharCoat JF can be used in:

- Offshore exploration and production
- Onshore gas and distribution
- Hydrocarbon and chemical processing
- Offshore rigs and platforms
- FPSO's
- Oil and gas terminals
- Refineries and petrochemical plants
- LNG Terminals
- Fuel tankers - road and rail cars



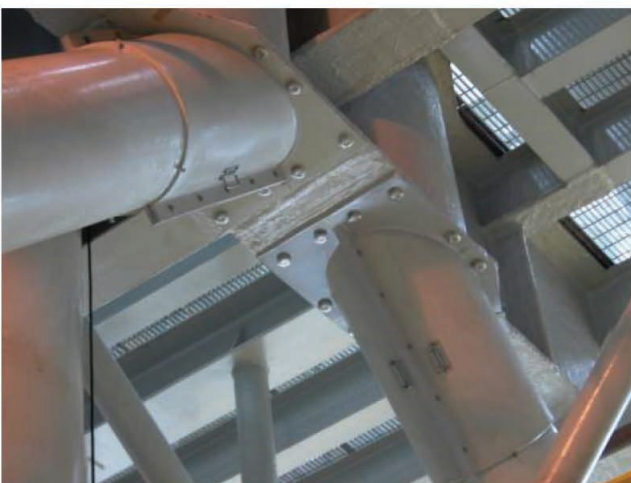
ADVANTAGES

- ISO 22899-1:2007 (E) Tested and Approved
- DOT - Federal Rail Admin Approved
- NORSOK Approval
- AS1530.8.1 and ENA Approved for Power Poles
- Non-Toxic Non-Hazardous
- Solvent Free
- Plural Component Spray Application
- Airless Spray application
- Cast Grade for Piping, Actuators, Pumps & Trays
- 3.25mm Single Coat (2-3 passes)
- Less than 3.5mm DFT for Vessel Protection under FRA
- Direct to metal application, no primer needed

CASTING CHARCOAT JF

For valves, actuators, piping and other non-movable machinery and parts, CharCoat JF has a solution through technology! Typical of Lockwell Systems coatings, we do not interrupt or shutdown the asset.

Utilizing 3D scanning technology, casting of machinery is completed off-site, while the asset stays on-site and online. After casting of the unit is complete, our specialists fit the cast on-site, creating access panels and piping outlets in minimal time without interfering with the units capacity.



SPRAYING CHARCOAT JF

Spraying CharCoat JF can be completed with either a standard airless unit, or plural component machines.

With standard airless units, once mixed CharCoat JF has a 40 minute pot life and is ideal for small areas where portable machinery is required.

For larger projects, we recommend a complete plural component system.



PROPERTY	TEST METHOD	VALUE
Liquid Density		
Part A	ASTM D1475	≥1.23 g/cc, 10.3lbs/gal
Part B	ASTM D1475	≥1.14 g/cc, 9.5lbs/gal
Liquid Mixed Density	ASTM D1475	≥1.20 g/cc, 10.0lbs/gal
Typical Sprayed Density	ASTM D1475	≥1.10 g/cc
Moisture Absorption	Immersion in water	
	At RT for 31 days - uncoated	2.6 wt % weight gain
Flame Spread	ASTM E 84	25
Smoke Generation	ASTM E 84	90
Hardness	Shore D	≥72
Chemical Resistance	Resistant to many common acids, bases and solvents	
Tensile Strength	ASTM D638	≥1800 psi
Tensile Modulus	ASTM D638	≥200 K psi
Compression Strength	ASTM D695	≥2500 psi
Lap Shear Strength	ASTM D1002	≥1400 psi
Flexible Strength	ASTM D790	≥2800 psi
Flexural Modulus	ASTM D790	≥210 K psi
Notched IZOD Impact Strength	ASTM D256	0.63 J/cm
Smoke & Toxicity Index	IMO FTP Code Annex.1, Pt2	Toxicity Pass Smoke Primary Deck/floor Pass
Torch & Hose Stream	NFPA 58	120 minutes
Federal Rail Admin (FRA)	49CFR & 179.18	Less than 3.5mm

Disclaimer: The above data, particularly the recommendations for the application and use of **CharCoat Passive Fire Protection** products are based on the manufacturer's knowledge and experience. Due to different materials and conditions of application, which are beyond our control, we recommend in any case to carry out sufficient tests in order to ensure that **CharCoat Passive Fire Protection** products are suitable for the intended purpose and applications. Therefore, any liability for such recommendations or any oral advice is expressly excluded unless we have acted willfully or by gross negligence. It is always the responsibility of the installer / purchaser to guarantee correct preparation, DFT (CharCoat Coatings) and thickness (CharCoat Firestop Products) of all **CharCoat Passive Fire Protection** products. **CharCoat Passive Fire Protection** is not liable for installation or faulty installation. It is always the responsibility of the installer / purchaser to guarantee and certify the installation of materials.

CHARCOAT

Suite 301 – 220 Brew Street, Port Moody, BC V3H 0H6 Canada

+1 604 941 1001 mail@CharCoat.com www.CharCoat.com

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AUTHOR	ROMNICK YUMOL
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